

Unit 10

GROUP 2

58 items · 164 marks

Syllabus sections in this unit

10.1 Similarities and trends in the properties of the Group 2 metals, magnesium to barium, and their compounds

Unit 10: Group 2**1. M/J 26/P24/Q3(c)**

(c) (i) Describe what is observed when white solid strontium nitrate decomposes on heating.

[1]

(ii) State the trend in solubility down the Group 2 hydroxides.

[1]

2. MAR 26/P22/Q1(b)

Context

1 The Group 2 elements Mg to Ba show trends in their physical properties down the group.

(b) When strontium reacts with water, an alkaline solution is produced.

(i) Write an equation for this reaction. Include state symbols.

[2]

(ii) Samples of the alkaline solution produced in 1(b)(i) are tested with dilute HCl(aq) and separately with dilute $\text{H}_2\text{SO}_4\text{(aq)}$.

State what you would observe in each test.

observation with dilute HCl(aq)

observation with dilute $\text{H}_2\text{SO}_4\text{(aq)}$

[2]

3. MAR 26/P22/Q1(c)

Context

1 The Group 2 elements Mg to Ba show trends in their physical properties down the group.

(c) The nitrates and carbonates of Group 2 elements decompose on heating.

(i) State the trend in the thermal stability of the Group 2 carbonates down the group.

[1]

(ii) Write an equation for the thermal decomposition of $\text{Ca(NO}_3)_2$.

[1]

4. O/N 25/P21/Q2(d)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(d) Group 2 nitrates decompose on heating to form oxides.

(i) State the trend in thermal stability of the Group 2 nitrates down the group.

[1]

(ii) Identify the other products of the thermal decomposition of Group 2 nitrates.

[1]

5. O/N 25/P22/Q2(f)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(f) Aqueous MgCl_2 reacts with aqueous Na_2CO_3 to form a white precipitate. Upon heating, the white precipitate undergoes thermal decomposition.

(i) Construct an equation to show the reaction of aqueous MgCl_2 with aqueous Na_2CO_3 . Use state symbols in your equation.

[2]

(ii) Identify the products of the thermal decomposition of the white precipitate.

[1]

(iii) State the trend in thermal stability of the Group 2 carbonates down the group.

[1]

6. O/N 25/P23/Q2(d)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(d) Group 2 nitrates decompose on heating to form oxides.

(i) State the trend in thermal stability of the Group 2 nitrates down the group.

[1]

(ii) Identify the other products of the thermal decomposition of Group 2 nitrates.

[1]

7. MAR 25/P22/Q2(d)

(d) (i) State what is observed when dilute hydrochloric acid is added to separate samples of barium oxide and barium carbonate.

barium oxide

barium carbonate

[2]

(ii) Write an equation for the reaction of strontium, Sr, with an excess of cold water.

[1]

(iii) State the variation in solubility of the Group 2 sulfates down the group.

[1]

8. O/N 24/P21/Q3(b)(i)-(ii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (b) (introduction)

(b) Carbon is found in inorganic compounds such as carbonates.

(i) Write an equation for the reaction of magnesium carbonate with dilute $\text{HCl}(\text{aq})$.

[1]

(ii) Describe the thermal stability of the carbonates down Group 2.

[1]

9. O/N 24/P22/Q3(a)(iv)-(v)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

Context: part (a) (introduction)

(a) Nitrogen is found in inorganic compounds such as nitrogen oxides (NO_x), nitrates and nitric acid.

(iv) Nitric acid reacts with basic oxides to form nitrates.

Write an equation for the reaction of nitric acid with calcium oxide.

[1]

(v) Describe what is seen when solid calcium nitrate is heated strongly.

[1]

10. O/N 24/P23/Q3(b)(i)-(ii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (b) (introduction)

(b) Carbon is found in inorganic compounds such as carbonates.

(i) Write an equation for the reaction of magnesium carbonate with dilute $\text{HCl}(\text{aq})$.

[1]

(ii) Describe the thermal stability of the carbonates down Group 2.

[1]

11. M/J 24/P22/Q2(c)

Context

2 Separate samples of Na_2CO_3 and NaHCO_3 react with $\text{HCl}(\text{aq})$ to produce the same products, as shown in Table 2.1.

Table 2.1

reaction	equation	$\Delta H/\text{kJ mol}^{-1}$
1	$\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$	ΔH_1
2	$\text{NaHCO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$	$\Delta H_2 = +27.2$

(c) **Z** is a salt that contains a Period 4 element from Group 2. When **Z** is heated brown gas forms.

Identify the formula of **Z** and use it to write an equation for the reaction.

[2]

12. M/J 24/P23/Q1(d)(ii)

Context: part (d) (introduction)

(d) The oxide of silicon reacts with calcium oxide in an addition reaction to produce calcium silicate, CaSiO_3 . The oxidation number of calcium in CaSiO_3 is +II.

(ii) Calcium oxide can be made from calcium carbonate in a single-step reaction.

Identify the type of reaction that occurs.

[1]

13. O/N 23/P21/Q2(a)

Context

2 Barium hydroxide, $\text{Ba}(\text{OH})_2$, is a strong base used in inorganic and organic reactions.

Fig. 2.1 shows a reaction scheme involving $\text{Ba}(\text{OH})_2$.

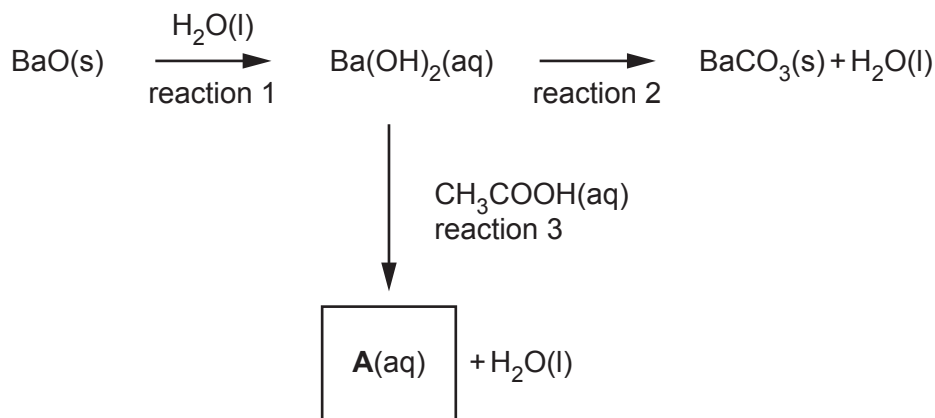


Fig. 2.1

(a) (i) State the variation in solubilities of group 2 hydroxides.

[1]

(ii) State what is observed in reaction 1.

[1]

(iii) Suggest a reactant for reaction 2.

[1]

(iv) Identify **A**.

[1]

(v) $\text{Ba}(\text{OH})_2$ is made by the reaction of Ba with water.

Write an equation for this reaction.

[1]

14. O/N 23/P21/Q2(b)

Context

2 Barium hydroxide, $\text{Ba}(\text{OH})_2$, is a strong base used in inorganic and organic reactions.

Fig. 2.1 shows a reaction scheme involving $\text{Ba}(\text{OH})_2$.

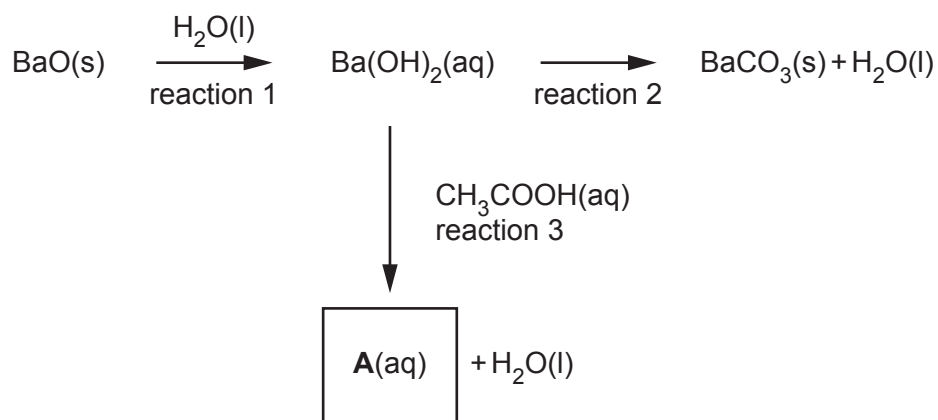


Fig. 2.1

(b) The mineral barytocalcite contains both BaCO_3 and CaCO_3 . Both compounds decompose on heating.

(i) State which compound decomposes first when barytocalcite is heated.

Explain your answer.

[1]

(ii) Construct an equation for the complete thermal decomposition of barytocalcite.

The formula of barytocalcite is $\text{BaCa}(\text{CO}_3)_2$.

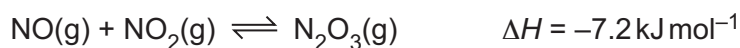
$\text{BaCa}(\text{CO}_3)_2$

[1]

15. O/N 23/P22/Q2(f)

Context

- 2** NO and NO₂ react at 25 °C to give N₂O₃ as shown in the equation.



The reaction is reversible and reaches equilibrium in a closed system.

- (f)** A student titrates nitric acid with a base to form a solution containing aqueous magnesium nitrate.

- (i)** Identify a base that the student could use.

[1]

- (ii)** The student evaporates the water to obtain magnesium nitrate solid. When this solid is heated it decomposes.

Write an equation for the decomposition of magnesium nitrate.

[1]

- (iii)** State how the thermal stability of Group 2 nitrates changes down the group.

[1]

16. O/N 23/P23/Q2(a)

Context

2 Barium hydroxide, $\text{Ba}(\text{OH})_2$, is a strong base used in inorganic and organic reactions.

Fig. 2.1 shows a reaction scheme involving $\text{Ba}(\text{OH})_2$.

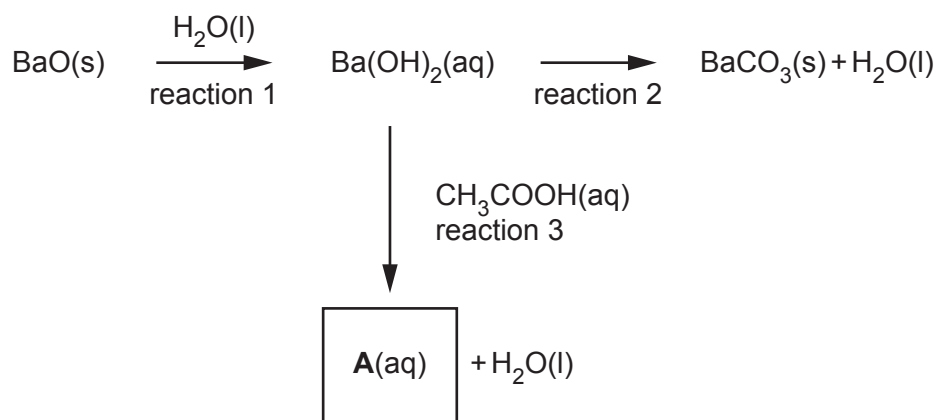


Fig. 2.1

(a) (i) State the variation in solubilities of group 2 hydroxides.

[1]

(ii) State what is observed in reaction 1.

[1]

(iii) Suggest a reactant for reaction 2.

[1]

(iv) Identify **A**.

[1]

(v) $\text{Ba}(\text{OH})_2$ is made by the reaction of Ba with water.

Write an equation for this reaction.

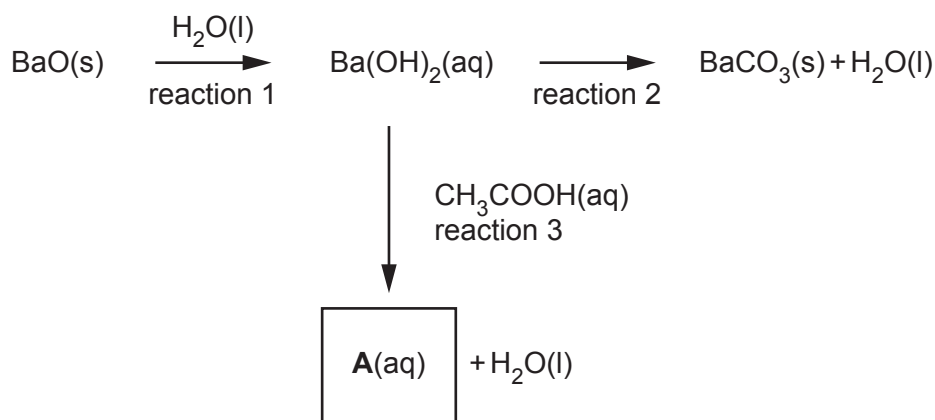
[1]

17. O/N 23/P23/Q2(b)

Context

- 2** Barium hydroxide, $\text{Ba}(\text{OH})_2$, is a strong base used in inorganic and organic reactions.

Fig. 2.1 shows a reaction scheme involving $\text{Ba}(\text{OH})_2$.

**Fig. 2.1**

- (b)** The mineral barytocalcite contains both BaCO_3 and CaCO_3 . Both compounds decompose on heating.

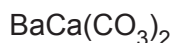
- (i)** State which compound decomposes first when barytocalcite is heated.

Explain your answer.

[1]

- (ii)** Construct an equation for the complete thermal decomposition of barytocalcite.

The formula of barytocalcite is $\text{BaCa}(\text{CO}_3)_2$.

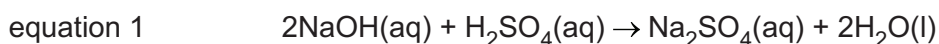


[1]

18. M/J 23/P21/Q2(c)(i)

Context

- 2** A neutralisation reaction occurs when NaOH(aq) is added to $\text{H}_2\text{SO}_4\text{(aq)}$.



- (c) (i)** Complete the equation for the reaction that occurs when a solution of $\text{Ba}(\text{OH})_2$ is added to aqueous sulfuric acid. Include state symbols.



[4]

19. M/J 23/P23/Q2(b)

- (b) Aluminium nitrate is a white soluble salt. On heating aluminium nitrate, thermal decomposition occurs and a brown gas is seen.

State the formula of the salt of another element in Period 3 which also decomposes on heating to produce a brown gas.

[1]

20. MAR 23/P22/Q1(d)(iii)

Context

- 1 The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy/kJ mol ⁻¹	1310	519	1090	1310	1000
second ionisation energy/kJ mol ⁻¹	—	7300	2350	3390	2260

Context: part (d) (introduction)

- (d) CO₂ and SO₂ are acidic gases.

(iii) Construct an equation for the reaction of CO₂ with Mg(OH)₂.

[1]

21. MAR 23/P22/Q2(b)

Context

- 2 The Group 2 elements Mg to Ba are all silvery-white reactive metals.

(b) Write an equation for the reaction of magnesium with cold water.

[1]

22. MAR 23/P22/Q2(c)

Context

- 2 The Group 2 elements Mg to Ba are all silvery-white reactive metals.

(c) Identify a single reagent that can be used to distinguish separate samples of dilute Mg(NO₃)₂(aq) and dilute Ba(NO₃)₂(aq).

Explain your answer.

reagent

explanation

[2]

23. O/N 22/P21/Q2(b)

Context

2 Magnesium shows reactions typical of a Group 2 metal.

(b) Fig. 2.1 shows some reactions of magnesium and its compounds.

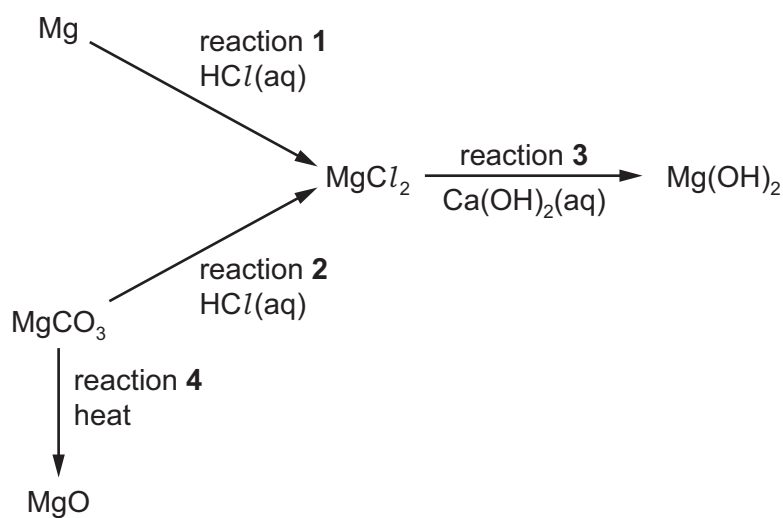


Fig. 2.1

(i) Identify the other products of reactions **1** and **2**.

reaction **1**

reaction **2**

[2]

(ii) Reaction **3** is used to form a precipitate of Mg(OH)₂ from MgCl₂(aq).

State why Ca(OH)₂(aq) would **not** form a precipitate of Ba(OH)₂ from BaCl₂(aq).

[1]

(iii) State the type of reaction that occurs in reaction **4**.

[1]

24. O/N 22/P23/Q2(b)

Context

2 Magnesium shows reactions typical of a Group 2 metal.

(b) Fig. 2.1 shows some reactions of magnesium and its compounds.

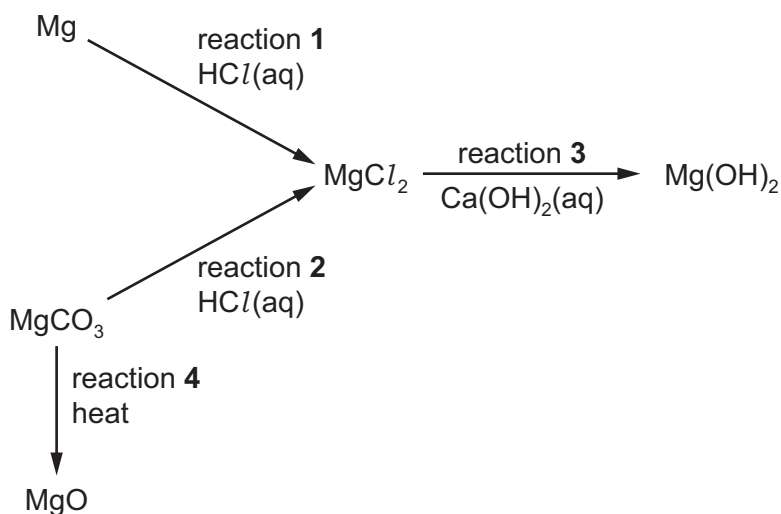


Fig. 2.1

(i) Identify the other products of reactions **1** and **2**.

reaction **1**

reaction **2**

[2]

(ii) Reaction **3** is used to form a precipitate of Mg(OH)₂ from MgCl₂(aq).

State why Ca(OH)₂(aq) would **not** form a precipitate of Ba(OH)₂ from BaCl₂(aq).

[1]

(iii) State the type of reaction that occurs in reaction **4**.

[1]

25. M/J 22/P21/Q1(b)

Context

1 Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.

(b) (i) Write the equation for the thermal decomposition of calcium nitrate.

[1]

(ii) Suggest which of the Group 2 nitrates, calcium, magnesium or radium, requires the highest temperature to decompose. Explain your answer.

[1]

26. M/J 22/P21/Q1(c)

Context

- 1** Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.

(c) Predict what you would observe when aqueous radium chloride is added to aqueous sodium sulfate.

Do **not** refer to temperature changes in your answer.

[1]

27. M/J 22/P22/Q2(c)

Context

- 2** Radium, Ra, is an element found in Group 2 of the Periodic Table. It is a crystalline solid at room temperature and conducts electricity.

Radium chloride, RaCl_2 , has a melting point of 900°C and is soluble in water.

(c) Solid Ra and Ca show similar reactions with H_2O , but the reactions occur at different rates.

Separate samples, each containing a single piece of solid Ra or Ca, are added to equal volumes of cold water.

Each sample contains equal numbers of moles of solid and the H_2O is in excess.

(i) Construct an equation for the reaction of Ra with H_2O .

[1]

(ii) Identify which element, Ra or Ca, reacts with H_2O at a faster rate. Suggest how the observations of each reaction would differ.

[1]

(iii) Suggest why these reactions occur at different rates.

[2]

(iv) One of the solutions is cloudy when the reaction has finished.

At the end of each reaction, universal indicator is added to each reaction mixture.

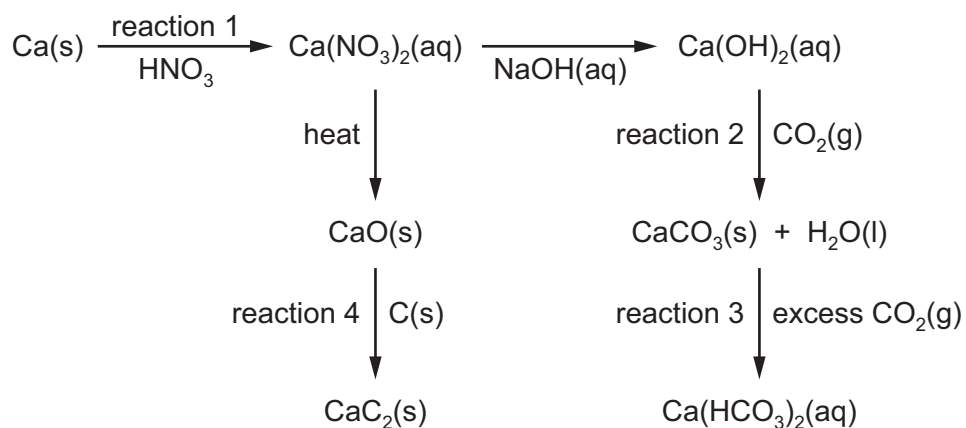
Suggest pH values of the solutions made in both reactions. Explain your answer.

[2]

28. O/N 21/P21/Q2(a)(i)-(iii)

Context

2 The reaction scheme shows some reactions of calcium.



(a) (i) Reaction 1 produces $\text{Ca(NO}_3)_2$ and one other product.

Identify the other product.

[1]

(ii) Construct an equation for the thermal decomposition of $\text{Ca(NO}_3)_2\text{(s)}$.

[1]

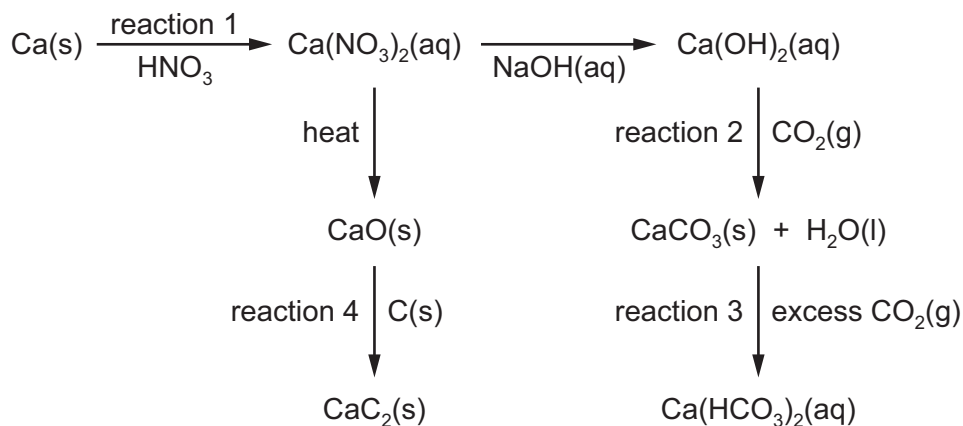
(iii) State the trend in the thermal stability of the Group 2 nitrates down the group.

[1]

29. O/N 21/P23/Q2(a)(i)-(iii)

Context

2 The reaction scheme shows some reactions of calcium.



(a) (i) Reaction 1 produces $\text{Ca(NO}_3)_2$ and one other product.

Identify the other product.

[1]

(ii) Construct an equation for the thermal decomposition of $\text{Ca(NO}_3)_2\text{(s)}$.

[1]

(iii) State the trend in the thermal stability of the Group 2 nitrates down the group.

[1]

30. M/J 21/P21/Q1(b)(ii)-(iii)

Context

1 Ethanedioic acid, $\text{HO}_2\text{CCO}_2\text{H}$, has a relative molecular mass of 90.0.

Context: part (b) (introduction)

(b) Solid ethanedioic acid reacts with aqueous calcium ions to make a precipitate of calcium ethanedioate, CaC_2O_4 .

CaC_2O_4 breaks down when heated to form calcium oxide, carbon dioxide and carbon monoxide.

(ii) Identify the type of reaction which occurs when CaC_2O_4 is heated.

[1]

(iii) Identify another compound containing calcium ions which will also produce carbon dioxide and calcium oxide when it is heated.

[1]

31. M/J 21/P22/Q1(f)(i)

Context

- 1** A Group 2 metal combines with bromine to form a crystalline solid, MBr_2 .

Excess aqueous AgNO_3 is added to a solution of MBr_2 and a precipitate forms. The mixture is filtered. The precipitate is dried and the mass of the precipitate is recorded.

Context: part (f) (introduction)

- (f)** Separate 1.0g samples of three different magnesium salts are tested in order to identify the anion present in each sample.
- (i)** Explain how the action of heat is used to identify which sample is:
- MgCO_3
 - $\text{Mg}(\text{NO}_3)_2$
 - MgO .

[3]

32. M/J 21/P22/Q1(g)

Context

- 1** A Group 2 metal combines with bromine to form a crystalline solid, MBr_2 .

Excess aqueous AgNO_3 is added to a solution of MBr_2 and a precipitate forms. The mixture is filtered. The precipitate is dried and the mass of the precipitate is recorded.

- (g)** A sample of $\text{MgCO}_3(\text{s})$ is distinguished from a sample of $\text{Mg}(\text{OH})_2(\text{s})$ by adding a small amount of each solid to $\text{HCl}(\text{aq})$.

State **one** similarity and **one** difference in these two reactions.

similarity

difference

[2]

33. O/N 20/P21/Q2(c)(i)

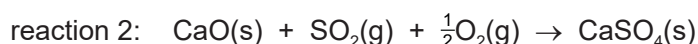
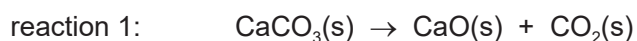
Context

- 2** Phosphorus, sulfur and chlorine can all react with oxygen to form oxides.

Context: part (c) (introduction)

- (c)** Emissions of SO_2 from coal-fired power stations can be reduced by mixing the coal with powdered limestone.

Limestone is heated to form CaO in reaction 1. This then reacts with SO_2 and O_2 to form CaSO_4 in reaction 2.



- (i)** State the type of reaction occurring in reaction 1.

[1]

34. O/N 20/P23/Q2(c)(i)

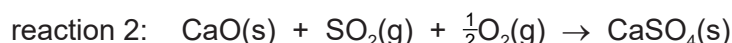
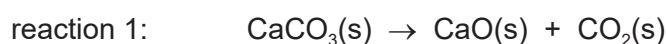
Context

2 Phosphorus, sulfur and chlorine can all react with oxygen to form oxides.

Context: part (c) (introduction)

(c) Emissions of SO_2 from coal-fired power stations can be reduced by mixing the coal with powdered limestone.

Limestone is heated to form CaO in reaction 1. This then reacts with SO_2 and O_2 to form CaSO_4 in reaction 2.



(i) State the type of reaction occurring in reaction 1.

[1]

35. MAR 20/P22/Q1(a)(i)

Context

1 Group 2 metals form alkaline solutions in water.

(a) (i) Write the equation for the reaction of calcium oxide with water.

[1]

36. O/N 19/P22/Q2(b)(ii)-(iv)

Context

2 Oxygen is the most abundant element in the Earth's crust. It reacts with other elements to form stable compounds, ions and molecules.

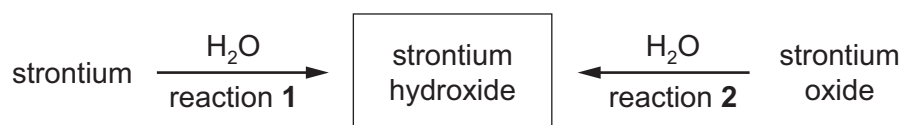
Context: part (b) (introduction)

(b) Group 2 elements form stable hydroxides, with general formula $\text{M}(\text{OH})_2$, where M is the Group 2 element.

(ii) Write an **ionic** equation for the reaction of magnesium hydroxide, $\text{Mg}(\text{OH})_2$, with hydrochloric acid.

[1]

(iii) Two methods of preparing strontium hydroxide are shown.



State **one** difference between the observations you would make for reaction 1 and reaction 2.

[1]

(iv) State how the solubility of the Group 2 hydroxides changes down the group.

[1]

37. M/J 19/P21/Q1(a)

1 (a) Group 2 elements share common chemical properties.

- (i) Calcium reacts in cold water more quickly than magnesium because more energy is required to remove the outer electrons in magnesium. This occurs even though calcium atoms have a greater nuclear charge.

Explain why more energy is required to remove the outer electrons in magnesium than in calcium.

[2]

- (ii) 0.001 mol of strontium reacts with an excess of cold water. When the reaction is complete a colourless solution is seen.

Construct the equation for the reaction of strontium with cold water. Include state symbols.

[2]

- (iii) 0.005 mol of calcium and 0.005 mol of strontium are added separately to two beakers. Each beaker contains 100 cm³ of cold water.

At the end of each reaction a white solid and a colourless solution are seen in both beakers.

Predict which element, calcium or strontium, produces the more alkaline solution. Explain your answer.

[2]

- (iv) Describe one observation when magnesium carbonate is added to excess dilute sulfuric acid.

[1]

38. M/J 19/P23/Q3(a)

3 (a) Magnesium reacts with oxygen to form magnesium oxide.

State **two** observations that would be made when magnesium is heated strongly and placed in a gas jar of pure oxygen.

[1]

39. M/J 19/P23/Q3(b)

also Unit 7 (1 mark); this unit 1 mark

- (b) Acid indigestion is caused by an excess of hydrochloric acid in the stomach. Magnesium oxide is commonly found in indigestion tablets.

(i) Give the equation to show how magnesium oxide relieves acid indigestion.

[1]

(ii) Name the type of reaction that occurs in (b)(i).

[1]

40. M/J 19/P23/Q3(d)

(d) Magnesium oxide can be made from magnesium carbonate in a one-step reaction using heat.

(i) Write an equation for this reaction. Include state symbols.

[1]

(ii) Name the type of reaction occurring during this process.

[1]

41. MAR 19/P22/Q1(b)(i)

Context

1 Nitrogen, N_2 , is the most abundant gas in the Earth's atmosphere and is very unreactive.

Context: part (b) (introduction)

(b) Magnesium and lithium both form nitrides with N_2 . These compounds both contain the N^{3-} ion.

(i) Write an equation for the reaction of magnesium with N_2 to form magnesium nitride.

[1]

42. O/N 18/P21/Q2(d)(iii)-(iv)

Context: part (d) (introduction)

(d) Some soils have compounds such as ammonium nitrate, calcium carbonate and calcium hydroxide added to them.

(iii) Another fertiliser, calcium ammonium nitrate, is formed when solid calcium carbonate is added to a mixture of aqueous ammonium nitrate and dilute nitric acid.

Suggest what would be **observed** in this reaction.

[2]

(iv) Calcium nitrate decomposes at a higher temperature than calcium ammonium nitrate.

Write an equation for the thermal decomposition of calcium nitrate.

[1]

43. O/N 18/P23/Q2(d)(iii)-(iv)

Context: part (d) (introduction)

(d) Some soils have compounds such as ammonium nitrate, calcium carbonate and calcium hydroxide added to them.

(iii) Another fertiliser, calcium ammonium nitrate, is formed when solid calcium carbonate is added to a mixture of aqueous ammonium nitrate and dilute nitric acid.

Suggest what would be **observed** in this reaction.

[2]

(iv) Calcium nitrate decomposes at a higher temperature than calcium ammonium nitrate.

Write an equation for the thermal decomposition of calcium nitrate.

[1]

44. M/J 18/P22/Q1(a)

Context

1 Sodium and magnesium are the first two elements in the third period.

(a) Sodium and magnesium both react with cold water to produce the same type of product in solution. With sodium the solution is clear but with magnesium it appears cloudy.

(i) Write an equation for the reaction of magnesium with cold water.

[1]

(ii) Suggest why the solution is cloudy after the reaction of magnesium with cold water.

[1]

45. M/J 18/P22/Q1(b)(i)-(iii)

Context

1 Sodium and magnesium are the first two elements in the third period.

Context: part (b) (introduction)

(b) Group 2 elements, including magnesium, react with oxygen and with dilute acids. There are trends in both the physical and chemical properties of the elements and their compounds down the group. Reactivity generally increases from Mg to Ba.

(i) Explain why there is a general increase in reactivity from Mg to Ba.

[2]

(ii) Give **two** observations for the reaction of magnesium with oxygen. Write an equation for this reaction. Include state symbols.

equation

[3]

(iii) Write an equation for the reaction of magnesium with sulfuric acid.

[1]

46. MAR 18/P22/Q3(a)

Context

3 Calcium and its compounds have a large variety of applications.

(a) Calcium metal reacts readily with most acids.

- (i)** Write an equation for the reaction of calcium with dilute nitric acid. State symbols are **not** required.

[1]

- (ii)** When calcium metal is placed in dilute sulfuric acid, it reacts vigorously at first.

After a short time, a crust of calcium sulfate forms on the calcium metal and the reaction stops. Some of the calcium metal and dilute sulfuric acid remain unreacted.

Suggest an explanation for these observations.

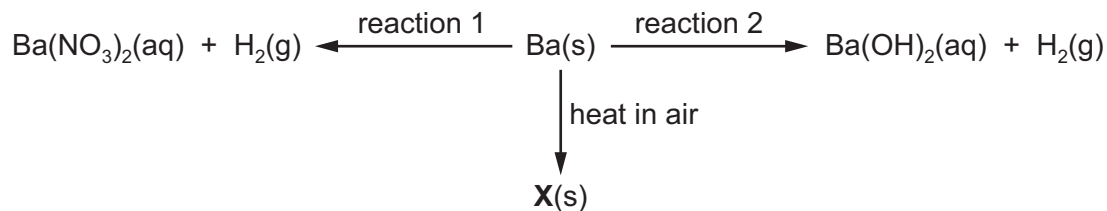
[2]

47. O/N 17/P22/Q3(c)

Context

- 3** The elements in Group 2 show trends in their properties that are typical of metals.
The elements in Group 17 show trends in their properties that are typical of non-metals.

(c) Some reactions based on the Group 2 metal barium, Ba, are shown.



- (i) State the reagent needed for each of reactions 1 and 2.

reaction 1

reaction 2

[2]

- (ii) Name **X** and write an equation for its formation.

name

equation

[2]

- (iii) The $\text{Ba(NO}_3)_2(\text{aq})$, produced by reaction 1, is heated to dryness. The anhydrous solid is then heated strongly and decomposes. Barium oxide is produced, together with two other products.

Identify the **two** other products of this decomposition reaction and state what would be observed.

[2]

- (iv) State what would be observed when excess $\text{MgSO}_4(\text{aq})$ is added to the $\text{Ba(OH)}_2(\text{aq})$ produced in reaction 2. Explain your answer.

[3]

48. M/J 17/P21/Q3(a)(iii)

Context

- 3** The properties of elements and their compounds show similarities, differences and trends depending on the positions of the elements in the Periodic Table.

Context: part (a) (introduction)

- (a) The positions of some elements are indicated. The letters used are **not** the symbols of the elements.

[illegible]

From the elements labelled, give the letter for;

- (iii) the element that forms a soluble hydroxide and an insoluble sulfate, [1]

49. M/J 17/P21/Q3(b)

Context

- 3** The properties of elements and their compounds show similarities, differences and trends depending on the positions of the elements in the Periodic Table.

- (b)** The elements in Group 2 all react with oxygen and with water.

- (i) State and explain the conditions needed for magnesium to react with oxygen.

[2]

- (ii) State what would be seen during the reaction in (b)(i).

[1]

- (iii) Write an equation for the reaction of magnesium with cold water. Include state symbols.

[2]

50. M/J 17/P21/Q3(c)

Context

- 3** The properties of elements and their compounds show similarities, differences and trends depending on the positions of the elements in the Periodic Table.

(c) The carbonates and nitrates of the elements in Group 2 can all be decomposed by heating.

- (i)** Write an equation for the thermal decomposition of magnesium nitrate.

[1]

- (ii)** The thermal decomposition of calcium carbonate forms a solid product that is industrially important. This solid product reacts with water to form a compound commonly known as slaked lime.

Write equations for the thermal decomposition of calcium carbonate and the reaction of the solid product to form slaked lime.

thermal decomposition

formation of slaked lime

[2]

51. M/J 17/P21/Q3(d)(ii)

Context

- 3** The properties of elements and their compounds show similarities, differences and trends depending on the positions of the elements in the Periodic Table.

Context: part (d) (introduction)

(d) Calcium carbonate and calcium hydroxide both have an important use in agriculture.

- (ii)** Write an ionic equation to illustrate this use of calcium carbonate.

[1]

52. M/J 16/P21/Q3(a)

Context

- 3** The elements in Group 2, and their compounds, show many similarities and trends in their properties.

(a) Magnesium, calcium, strontium and barium all react with cold water.

- (i)** Describe what you would **see** when some calcium is added to cold water.

[3]

- (ii)** Write an equation for the reaction taking place in **(i)**.

[1]

- (iii)** Describe how the reaction of barium with cold water would differ from the reaction of calcium in **(i)** in terms of what you would **see**.

[1]

53. M/J 16/P21/Q3(b)(iv)

Context

3 The elements in Group 2, and their compounds, show many similarities and trends in their properties.

Context: part (b) (introduction)

(b) Magnesium oxide can be formed by the reaction of magnesium and oxygen in the air.

(iv) Suggest a reason why magnesium oxide cannot be used as a lining for any furnaces containing acidic materials.

[1]

54. M/J 16/P21/Q3(c)

Context

3 The elements in Group 2, and their compounds, show many similarities and trends in their properties.

(c) The nitrates and carbonates of the Group 2 elements, from magnesium to barium, decompose when heated.

(i) State the trend in the temperature of thermal decomposition of these Group 2 nitrates and carbonates.

[1]

(ii) Give the equation for the thermal decomposition of magnesium carbonate.

[1]

(iii) Give the equation for the thermal decomposition of calcium nitrate.

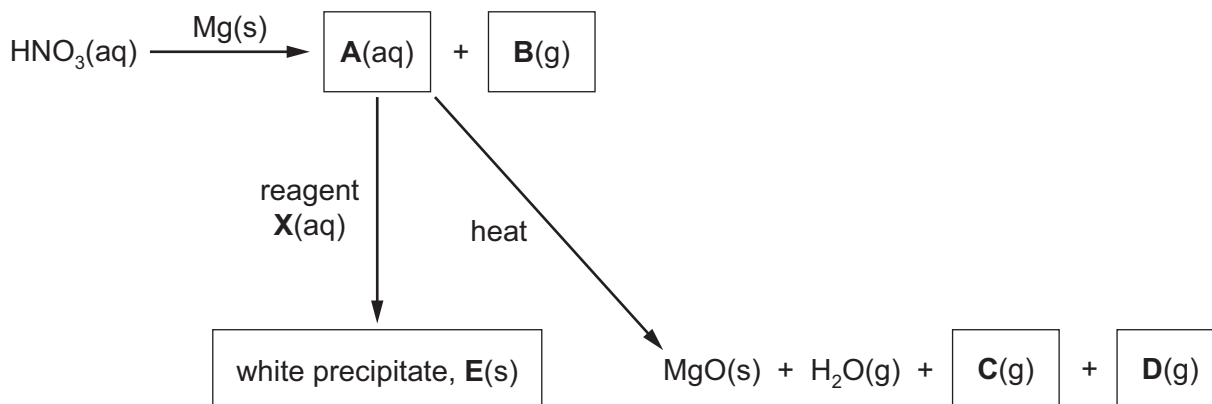
[1]

55. MAR 16/P22/Q1(b)

Context

1 This question is about Period 3 elements and their compounds.

(b) The flow chart below shows a series of reactions.



(i) Give the **formula** of each of the compounds **A** to **D**.

A**B****C****D**

[4]

(ii) **E** reacts with dilute aqueous acid to produce a gas that turns limewater cloudy.

Suggest the identity of reagent **X**.

[1]

56. M/J 15/P23/Q2(a)

Context

2 The elements in Group II, and their compounds, show a variety of trends in their properties.

(a) Magnesium, calcium and barium all react with cold water to form hydroxides.

(i) Describe and explain the trend in reactivity of these three elements with cold water.

[3]

(ii) Give the equation for the reaction of magnesium with cold water.

[1]

(iii) Suggest why the water eventually turns cloudy during the reaction of magnesium with cold water.

[1]

(iv) Suggest the equation for the reaction of hot magnesium with steam.

[1]

57. M/J 15/P23/Q2(b)(i)-(iii)

Context

2 The elements in Group II, and their compounds, show a variety of trends in their properties.

Context: part (b) (introduction)

(b) The oxides of magnesium, calcium and barium all react with dilute nitric acid to form nitrates.

(i) Give the equation for the reaction of magnesium oxide with nitric acid.

[1]

(ii) State the trend in thermal stability of the nitrates of Group II.

[1]

(iii) Give the equation for the thermal decomposition of magnesium nitrate.

[1]

58. M/J 15/P23/Q2(d)(i)-(ii)

Context

2 The elements in Group II, and their compounds, show a variety of trends in their properties.

Context: part (d) (introduction)

(d) The word 'lime' is usually used to refer to a range of calcium-containing compounds that have a range of uses.

(i) Write equations to show how calcium carbonate can be converted into calcium hydroxide by a two-step process.

[2]

A garden pond, with a total volume of 8000 dm^3 , has been contaminated in such a way that its pH has fallen to 4. This means that the concentration of hydrogen ions, H^+ , in the water is $1 \times 10^{-4} \text{ mol dm}^{-3}$.

(ii) Write an ionic equation for the neutralisation reaction that occurs between hydrogen ions and carbonate ions, CO_3^{2-} .

[1]

Answers Section

1. M/J 26/P24/Q3(c)

3(c)(i)	brown gas	1	gas = fumes = vapour A red-brown gas, orange-brown gas A shades of brown R red gas (alone) I white solid (formed) I any other observations (e.g. melting, effervescence, solid disappears)
3(c)(ii)	increases	1	A decrease <u>up</u> the group. A more soluble

2. MAR 26/P22/Q1(b)

1(b)(i)	M1 $\text{Sr(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{Sr(OH)}_2\text{(aq)} + \text{H}_2\text{(g)}$ M2 state symbols for <u>all</u> species	2
1(b)(ii)	M1 with HCl no visible reaction / no change (observed) M2 with H_2SO_4 white precipitate/solid forms	2

3. MAR 26/P22/Q1(c)

1(c)(i)	(thermal stability) increases (down the group)	1
1(c)(ii)	$2\text{Ca(NO}_3)_2 \rightarrow 2\text{CaO} + 4\text{NO}_2 + \text{O}_2$	1

4. O/N 25/P21/Q2(d)

2(d)(i)	(thermal stability) increases (down the group)	1
2(d)(ii)	NO_2 AND O_2	1

5. O/N 25/P22/Q2(f)

2(f)(i)	$\text{MgCl}_2\text{(aq)} + \text{Na}_2\text{CO}_3\text{(aq)} \rightarrow \text{MgCO}_3\text{(s)} + 2\text{NaCl(aq)}$ M1 formulae in balanced equation M2 state symbols	2
2(f)(ii)	MgO / magnesium oxide AND CO_2 / carbon dioxide	1
2(f)(iii)	(thermal stability) increases (down the group)	1

6. O/N 25/P23/Q2(d)

2(d)(i)	(thermal stability) increases (down the group)	1
2(d)(ii)	NO_2 AND O_2	1

7. MAR 25/P22/Q2(d)

2(d)(i)	<i>barium oxide</i> (white) solid disappears / dissolves <i>barium carbonate</i> effervescence AND (white) solid disappears / dissolves	2
2(d)(ii)	$\text{Sr} + 2\text{H}_2\text{O} \rightarrow \text{Sr(OH)}_2 + \text{H}_2$	1
2(d)(iii)	decreases (down the group)	1

8. O/N 24/P21/Q3(b)(i)-(ii)

3(b)(i)	$\text{MgCO}_3 + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$	1
3(b)(ii)	thermal stability increases down the group	1

9. O/N 24/P22/Q3(a)(iv)-(v)

3(a)(iv)	$2\text{HNO}_3 + \text{CaO} \rightarrow \text{Ca}(\text{NO}_3)_2 + \text{H}_2\text{O}$	1
3(a)(v)	brown fumes given off	1

10. O/N 24/P23/Q3(b)(i)-(ii)

3(b)(i)	$\text{MgCO}_3 + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$	1
3(b)(ii)	thermal stability increases down the group	1

11. M/J 24/P22/Q2(c)

2(c)	$\text{Ca}(\text{NO}_3)_2 \rightarrow \text{CaO} + 2\text{NO}_2 + \frac{1}{2}\text{O}_2$ M1 correct formula for calcium nitrate: $\text{Ca}(\text{NO}_3)_2$ M2 correct balanced equation	2
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12. M/J 24/P23/Q1(d)(ii)

1(d)(ii)	thermal decomposition	1
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13. O/N 23/P21/Q2(a)

2(a)(i)	increases down the group	1
2(a)(ii)	white solid disappears	1
2(a)(iii)	CO_2 / carbon dioxide	1
2(a)(iv)	$\text{Ba}(\text{CH}_3\text{COO})_2$ / barium ethanoate	1
2(a)(v)	$\text{Ba} + 2\text{H}_2\text{O} \rightarrow \text{Ba}(\text{OH})_2 + \text{H}_2$	1

14. O/N 23/P21/Q2(b)

2(b)(i)	CaCO_3 AND thermal stability of carbonates increases down the group	1
2(b)(ii)	$\text{BaCa}(\text{CO}_3)_2 \rightarrow \text{BaO} + \text{CaO} + 2\text{CO}_2$	1

15. O/N 23/P22/Q2(f)

2(f)(i)	magnesium hydroxide / $\text{Mg}(\text{OH})_2$	1
2(f)(ii)	$\text{Mg}(\text{NO}_3)_2 \rightarrow \text{MgO} + 2\text{NO}_2 + \frac{1}{2}\text{O}_2$	1
2(f)(iii)	increases	1

16. O/N 23/P23/Q2(a)

2(a)(i)	increases down the group	1
2(a)(ii)	white solid disappears	1
2(a)(iii)	CO_2 / carbon dioxide	1
2(a)(iv)	$\text{Ba}(\text{CH}_3\text{COO})_2$ / barium ethanoate	1

2(a)(v)	$\text{Ba} + 2\text{H}_2\text{O} \rightarrow \text{Ba}(\text{OH})_2 + \text{H}_2$	1
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17. O/N 23/P23/Q2(b)

2(b)(i)	CaCO_3 AND thermal stability of carbonates increases down the group	1
2(b)(ii)	$\text{BaCa}(\text{CO}_3)_2 \rightarrow \text{BaO} + \text{CaO} + 2\text{CO}_2$	1

18. M/J 23/P21/Q2(c)(i)

2(c)(i)	..(1)..$\text{H}_2\text{SO}_4(\text{aq}) + \text{..(1)..Ba}(\text{OH})_2(\text{aq}) \rightarrow \text{...(1)..BaSO}_4(\text{s}) + 2\text{H}_2\text{O}(\text{l})$..... M1 correct state symbols for both products M2 correctly balanced equation	2
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19. M/J 23/P23/Q2(b)

2(b)	$\text{Mg}(\text{NO}_3)_2$	1
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20. MAR 23/P22/Q1(d)(iii)

1(d)(iii)	$\text{CO}_2 + \text{Mg}(\text{OH})_2 \rightarrow \text{MgCO}_3 + \text{H}_2\text{O}$	1
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21. MAR 23/P22/Q2(b)

2(b)	$\text{Mg} + 2\text{H}_2\text{O} \rightarrow \text{Mg}(\text{OH})_2 + \text{H}_2$	1
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22. MAR 23/P22/Q2(c)

2(c)	reagent = any named/formula of soluble sulfate OR H_2SO_4 OR any named/formula of soluble hydroxide	1
	BaSO₄ insoluble (& MgSO₄ soluble) OR (Ba(OH)₂ soluble &) Mg(OH)₂ insoluble	1

23. O/N 22/P21/Q2(b)

2(b)(i)	reaction 1 = hydrogen / H_2	1
	reaction 2 = carbon dioxide / CO_2 AND water / H_2O	1
2(b)(ii)	$\text{Ba}(\text{OH})_2$ is soluble (in aqueous solution) / solubility of Group 2 hydroxides increases down group	1
2(b)(iii)	thermal decomposition	1

24. O/N 22/P23/Q2(b)

2(b)(i)	reaction 1 = hydrogen / H_2	1
	reaction 2 = carbon dioxide / CO_2 AND water / H_2O	1
2(b)(ii)	$\text{Ba}(\text{OH})_2$ is soluble (in aqueous solution) / solubility of Group 2 hydroxides increases down group	1
2(b)(iii)	thermal decomposition	1

25. M/J 22/P21/Q1(b)

1(b)(i)	$\text{Ca}(\text{NO}_3)_2 \rightarrow \text{CaO} + 2\text{NO}_2 + \frac{1}{2}\text{O}_2$	1
1(b)(ii)	radium (nitrate) as thermal stability increases down group / has the greatest thermal stability	1

26. M/J 22/P21/Q1(c)

1(c)	white precipitate / solid (of radium sulfate)	1
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27. M/J 22/P22/Q2(c)

2(c)(i)	$\text{Ra} + 2\text{H}_2\text{O} \rightarrow \text{Ra}(\text{OH})_2 + \text{H}_2$	1
2(c)(ii)	Ra – more bubbles per unit time OR With Ra solid disappears more quickly OR Ra is the first to stop fizzing ora	1
2(c)(iii)	option 1 <i>suggest why these reactions occur at different rates (collision theory)</i> M1 difference in activation energy / ionisation energy M2 affects the frequency of effective collisions (between particles / molecules) OR affects the proportion of particles with energy greater than activation energy option 2 <i>suggest why the elements have different reactivity in terms of atomic structure (which results in a different rate)</i> M1 ionisation energies are different M2 a decrease in nuclear attraction due to EITHER increase in shielding OR increase in distance of outer electron from nucleus OR increase in number of shells of electrons ora	2
2(c)(iv)	M1 - pH value (or values) for each solution must be greater > 7 AND - pH value (or range values) identified for solution made from Ra must be greater than pH values (or range of values) stated for Ca M2 any one from: - solubility of group 2 hydroxides increases down the group $\text{Ra}(\text{OH})_2$ is more soluble - greater concentration of $\text{OH}^-(\text{aq})$ in the solution involving Ra - more $\text{OH}^-(\text{aq})$ in the solution involving Ra	2

28. O/N 21/P21/Q2(a)(i)-(iii)

2(a)(i)	hydrogen / H_2	1
2(a)(ii)	$\text{Ca}(\text{NO}_3)_2 \rightarrow \text{CaO} + 2\text{NO}_2 + \frac{1}{2}\text{O}_2$	1
2(a)(iii)	(thermal stability) increases	1

29. O/N 21/P23/Q2(a)(i)-(iii)

2(a)(i)	hydrogen / H_2	1
2(a)(ii)	$\text{Ca}(\text{NO}_3)_2 \rightarrow \text{CaO} + 2\text{NO}_2 + \frac{1}{2}\text{O}_2$	1
2(a)(iii)	(thermal stability) increases	1

30. M/J 21/P21/Q1(b)(ii)-(iii)

1(b)(ii)	(thermal) decomposition OR disproportionation	1
1(b)(iii)	calcium carbonate / CaCO_3	1

31. M/J 21/P22/Q1(f)(i)

1(f)(i)	<i>explain how the action of heat is used to identify the 3 samples</i>	1
	M1 nitrate AND carbonate lose mass / less than 1 g	
	M2 nitrate produces brown (NO ₂) fumes	1
	M3 MgO no reaction / MgO no change	1

32. M/J 21/P22/Q1(g)

1(g)	<i>One similarity</i> M1 solid(s) disappear(s)	1
	<i>One difference</i> M2 MgCO ₃ fizzes (due to CO ₂) OR no fizzing with Mg(OH) ₂	1

33. O/N 20/P21/Q2(c)(i)

2(c)(i)	thermal decomposition	1
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34. O/N 20/P23/Q2(c)(i)

2(c)(i)	thermal decomposition	1
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35. MAR 20/P22/Q1(a)(i)

1(a)(i)	CaO + H ₂ O → Ca(OH) ₂	1
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36. O/N 19/P22/Q2(b)(ii)-(iv)

2(b)(ii)	OH ⁻ + H ⁺ → H ₂ O	1
2(b)(iii)	reaction with strontium / reaction 1 will effervesce / fizz / bubble OR no fizzes / bubbles / effervescence with SrO	1
2(b)(iv)	increases	1

37. M/J 19/P21/Q1(a)

1(a)(i)	M1 (one) fewer (inner) shell of electrons / less shielding (effect) <i>ORA</i>	1
	M2 smaller distance of the outer electrons (from the nucleus) / stronger nuclear attraction to the (outer) electrons <i>ORA</i>	1
1(a)(ii)	Sr(s) + 2H ₂ O(l) → Sr(OH) ₂ (<u>aq</u>) + H ₂ (g)	1
	M1 species AND balancing	
	M2 state symbols	1
1(a)(iii)	M1 strontium AND forms a more soluble hydroxide	1
	M2 strontium hydroxide is a stronger base / produces more OH ⁻ / it dissociates more	1
1(a)(iv)	(white) solid dissolves / effervescence	1

38. M/J 19/P23/Q3(a)

3(a)	white light / flame AND (produces a) white / grey solid / ash / powder / smoke	1
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39. M/J 19/P23/Q3(b)

3(b)(i)	MgO + 2HCl → MgCl ₂ + H ₂ O	1
3(b)(ii)	neutralisation	1

40. M/J 19/P23/Q3(d)

3(d)(i)	$\text{MgCO}_3(\text{s}) \rightarrow \text{MgO}(\text{s}) + \text{CO}_2(\text{g})$	1
3(d)(ii)	(thermal) decomposition	1

41. MAR 19/P22/Q1(b)(i)

1(b)(i)	$3\text{Mg} + \text{N}_2 \rightarrow \text{Mg}_3\text{N}_2$	1
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42. O/N 18/P21/Q2(d)(iii)-(iv)

2(d)(iii)	M1 effervescence / fizzing / bubbling M2 solid disappears	2
2(d)(iv)	$2\text{Ca}(\text{NO}_3)_2 \rightarrow 2\text{CaO} + 4\text{NO}_2 + \text{O}_2$	1

43. O/N 18/P23/Q2(d)(iii)-(iv)

2(d)(iii)	M1 effervescence / fizzing / bubbling M2 solid disappears	2
2(d)(iv)	$2\text{Ca}(\text{NO}_3)_2 \rightarrow 2\text{CaO} + 4\text{NO}_2 + \text{O}_2$	1

44. M/J 18/P22/Q1(a)

1(a)(i)	$\text{Mg} + 2\text{H}_2\text{O} \rightarrow \text{Mg}(\text{OH})_2 + \text{H}_2$	1
1(a)(ii)	the product / $\text{Mg}(\text{OH})_2$ sparingly soluble / slightly soluble / less soluble (than NaOH)	1

45. M/J 18/P22/Q1(b)(i)-(iii)

1(b)(i)	any two from: (from Mg to Ba) larger (atomic) radius / more shells / more shielding - electron(s) are less tightly held (by nucleus) / less attracted (to nucleus) - ionisation energy / ies decrease(s) / electron(s) lost (more) easily (down the group) - lower activation energy (for the reactions down the group)	2
1(b)(ii)	white light / white flame	1
	white smoke / white solid	1
	$2\text{Mg}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{MgO}(\text{s})$	1
1(b)(iii)	$\text{Mg} + \text{H}_2\text{SO}_4 \rightarrow \text{MgSO}_4 + \text{H}_2$	1

46. MAR 18/P22/Q3(a)

3(a)(i)	$\text{Ca} + 2\text{HNO}_3 \rightarrow \text{Ca}(\text{NO}_3)_2 + \text{H}_2$	1
3(a)(ii)	CaSO_4 does not react (with sulfuric acid) coating / crust / protective layer / CaSO_4 prevents reaction (of sulfuric acid) with calcium	2

47. O/N 17/P22/Q3(c)

3(c)(i)	reaction 1: HNO_3 or nitric(V) acid	1
	reaction 2: water / H_2O	1
3(c)(ii)	barium oxide	1
	$2\text{Ba} + \text{O}_2 \rightarrow 2\text{BaO}$	1

3(c)(iii)	NO ₂ / nitrogen dioxide / nitrogen(IV) oxide AND O ₂ / oxygen	1
	(red / yellow-)brown gas OR gas given off that relights glowing splint	1
3(c)(iv)	<u>white</u> ppt / solid / suspension	1
	of BaSO ₄ / barium sulfate OR Mg(OH) ₂ / magnesium hydroxide	1
	BaSO ₄ is insoluble OR Mg(OH) ₂ is insoluble / partially / slightly / sparingly soluble	1

48. M/J 17/P21/Q3(a)(iii)

3(a)(iii)	G	1
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49. M/J 17/P21/Q3(b)

3(b)(i)	(strong) heating	1
	(to provide / overcome) <u>high</u> activation energy	1
3(b)(ii)	<u>white</u> flame / <u>white</u> light / <u>white</u> smoke / <u>white solid</u>	1
3(b)(iii)	Mg(s) + 2H ₂ O(l) → Mg(OH) ₂ (s) + H ₂ (g)	2

50. M/J 17/P21/Q3(c)

3(c)(i)	2Mg(NO ₃) ₂ → 2MgO + 4NO ₂ + O ₂	1
3(c)(ii)	CaCO ₃ → CaO + CO ₂	1
	CaO + H ₂ O → Ca(OH) ₂	1

51. M/J 17/P21/Q3(d)(ii)

3(d)(ii)	CaCO ₃ + 2H ⁺ → Ca ²⁺ + CO ₂ + H ₂ O OR CaCO ₃ + 2H ⁺ → Ca ²⁺ + H ₂ CO ₃	1
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52. M/J 16/P21/Q3(a)

3 (a) (i)	bubbles / effervescence / fizzing	[1]	max [3]
	calcium gets smaller / disappears	[1]	
	water turns cloudy / milky	[1]	
	calcium sinks	[1]	
(ii)	Ca + 2H ₂ O → Ca(OH) ₂ + H ₂	[1]	[1]
(iii)	faster bubbling / disappearance of Ba OR no / less precipitate forms (owtte)	[1]	[1]

53. M/J 16/P21/Q3(b)(iv)

(iv)	MgO is basic / reacts with acid	[1]	[1]
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54. M/J 16/P21/Q3(c)

(c) (i)	increases (down the group)	[1]	[1]
(ii)	MgCO ₃ → MgO + CO ₂	[1]	[1]

(iii)	$2\text{Ca}(\text{NO}_3)_2 \rightarrow 2\text{CaO} + 4\text{NO}_2 + \text{O}_2$	[1]	[1]
			[15]

55. MAR 16/P22/Q1(b)

(b) (i)	A = $\text{Mg}(\text{NO}_3)_2$ B = H_2 C = NO_2 OR O_2 D = O_2 OR NO_2	[1] [1] [1] [1]	[4]
(ii)	any Group I carbonate OR ammonium carbonate	[1]	[1]
			[12]

56. M/J 15/P23/Q2(a)

2 (a) (i)	Reactivity increases down the group OR reference to observations that indicate trend Outer electrons lost more easily down group Due to increased distance/shielding of outer electrons from nucleus	[1] [1] [1]	[3]
(ii)	$\text{Mg} + 2\text{H}_2\text{O} \rightarrow \text{Mg}(\text{OH})_2 + \text{H}_2$	[1]	[1]
(iii)	Magnesium hydroxide sparingly soluble/insoluble	[1]	[1]
(iv)	$\text{Mg} + \text{H}_2\text{O} \rightarrow \text{MgO} + \text{H}_2$	[1]	[1]

57. M/J 15/P23/Q2(b)(i)-(iii)

(b) (i)	$\text{MgO} + 2\text{HNO}_3 \rightarrow \text{Mg}(\text{NO}_3)_2 + \text{H}_2\text{O}$	[1]	[1]
(ii)	(thermal stability) increases down the group	[1]	[1]
(iii)	$2\text{Mg}(\text{NO}_3)_2 \rightarrow 2\text{MgO} + 4\text{NO}_2 + \text{O}_2$	[1]	[1]

58. M/J 15/P23/Q2(d)(i)-(ii)

(d) (i)	$\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca}(\text{OH})_2$	[1] [1]	[2]
(ii)	$2\text{H}^+ + \text{CO}_3^{2-} \rightarrow \text{CO}_2 + \text{H}_2\text{O}$	[1]	[1]